Jingxin Zhang

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RESEARCH PROFILE

I am a Postdoctoral Research Associate at the Institute of Mechanics, Chinese Academy of Sciences (CAS), working at the interface of molecular spectroscopy, scientific computing, and computational spectral analysis. I hold a PhD in Physics and Astronomy from the UCL ExoMol group. My research focuses on connecting accurate molecular spectroscopy with the physical interpretation of complex spectra, particularly how spectral resolution, model assumptions, and molecular data quality affect the physical information extracted from spectra.

I am the developer of ExoMolHR, a high-resolution molecular spectroscopic database designed for line-resolved astronomical and laboratory spectroscopy, and PyExoCross, an open-source framework for large-scale molecular spectroscopic calculations and spectral modelling under LTE and non-LTE conditions. My expertise includes empirical and theoretical molecular line lists, spectroscopic uncertainties, absorption and emission cross sections, high-resolution spectroscopy, scalable scientific software, and machine learning approaches to multi-component spectral analysis and parameter inference.

Building on this foundation, I aim to extend my research towards exoplanet atmospheric modelling and inference, investigating how molecular spectroscopy and atmospheric model assumptions propagate into retrieved atmospheric properties and whether atmospheric interpretations remain physically consistent across independent wavelength ranges and spectral resolutions.

RESEARCH EXPERIENCE

Institute of Mechanics, Chinese Academy of Sciences (CAS) May 2025 – Present
Postdoctoral Research Associate *Beijing, China*

- Develop high-throughput spectral synthesis and machine learning methods for component identification and parameter inference in strongly blended atomic and molecular spectra.
- Investigate physics constrained methods that separate genuine spectral structure from model mismatch and support diagnosis of non-predefined spectral components.
- Extend PyExoCross based line list and cross-section workflows for large synthetic spectral datasets and uncertainty-aware modeling.

University College London (UCL), ExoMol Group Feb 2021 – Apr 2025
PhD Researcher *London, UK*

- Developed ExoMolHR, including data architecture, uncertainty based transition selection, standardised high-resolution products, and web tools for observational spectroscopy.
- Developed and released PyExoCross for partition functions, thermodynamic and radiative quantities, LTE/non-LTE stick spectra, and molecular/atomic cross sections from large line lists.
- Built reproducible pipelines for validation, filtering, conversion, and post-processing of multi-terabyte spectroscopic datasets.

EDUCATION

University College London Feb 2021 – Apr 2025
Ph.D. in Physics and Astronomy (ExoMol group)
Thesis: Development, maintenance, and post-processing tools for astrophysical spectroscopic databases
Supervisors: Prof. Jonathan Tennyson, Prof. Sergey Yurchenko

University College London Sep 2019 – Dec 2020
M.Sc. in Scientific Computing *Grade: Distinction*
Supervisors: Prof. Jonathan Tennyson, Prof. Sergey Yurchenko
Selected Coursework: Applied Bayesian Methods, Machine Learning with Big Data, High-Performance Computing, Research Software Engineering with Python, Research Computing with C++

University of Liverpool Sep 2017 – Jun 2019
B.Sc. in Mathematics *Grade: Upper Second-Class Honours*

Xi'an Jiaotong - Liverpool University (XJTLU) Sep 2015 – Jun 2017
B.Sc. in Applied Mathematics (2+2 Joint Programme)

RESEARCH CONTRIBUTIONS

ExoMolHR: High-Resolution Molecular Spectroscopic Database and Web Platform Website

- Built the relational database architecture and data ingestion pipelines for empirical high-resolution spectra.
- Developed algorithmic workflows for parsing, filtering, and cross matching quantum state transitions to extract low uncertainty data necessary for high-resolution exoplanet templates.
- Designed web utilities enabling atmospheric modelers to dynamically query and visualize high-resolution datasets.
- ExoMolHR data products have exceeded 60 000 downloads.

PyExoCross: Scalable Computational Spectroscopic Post-processing GitHub | PyPI | Documentation

- Designed an open-source Python framework enabling rapid conversion of massive line lists into temperature- and pressure-dependent cross sections and thermodynamic or radiative quantities.
- Implemented calculations of partition functions, specific heats, cooling functions, radiative lifetimes, oscillator strengths, stick spectra, and cross sections for LTE and non-LTE environments.
- Integrated high-performance computing (HPC) workflows to optimize thermal population and line profile calculations over millions of molecular transitions.
- The current development extends the framework towards flexible in-memory analysis, uncertainty-aware modeling, and efficient generation of large spectral datasets.
- PyExoCross has exceeded 14 000 PyPI downloads.

Machine Learning for Multi-component Spectral Analysis Ongoing

- Develop machine learning methods for component identification and physical-parameter inference from strongly blended atomic and molecular spectra.
- Investigate spectral identifiability across variations in temperature, composition, spectral resolution, and noise, using synthetic spectra to evaluate when individual components and parameters can be reliably constrained.
- Current work is extending these approaches towards physics-informed inference and explicit treatment of model and spectroscopic uncertainties.

PUBLICATIONS

First-author and Corresponding-author* papers

- **Jingxin Zhang***, Sergei N. Yurchenko, Jonathan Tennyson*. *PyExoCross 2.0: A Python Framework for LTE and non-LTE Spectra, Cross Sections, and Spectroscopic Post-processing of Atomic and Molecular Line Lists*. RASTI, rzag094 (2026).
- **Jingxin Zhang***, Christian Hill, Jonathan Tennyson, Sergei N. Yurchenko. *ExoMolHR: A Relational Database of Empirical High-resolution Molecular Spectra*. ApJS, **276**(2):67 (2025).
- **Jingxin Zhang***, Jonathan Tennyson*, Sergei N. Yurchenko. *PyExoCross: a Python program for generating spectra and cross-sections from molecular line lists*. RASTI, **3**, 257–287 (2024).
- **Jingxin Zhang***, Christian Hill, Sergei N. Yurchenko, Jonathan Tennyson*. *The ExoMolHR2026 High-Resolution Molecular Spectroscopic Database: Standardized Data Products and Web Tools.* Manuscript in preparation (2026).
- Qingyang Liu, **Jingxin Zhang***, Haohua Duan*, Ramin Yahyapour*. *MET-C3: Mixed Exogenous–Temporal Dependency Modeling for Non-Stationary Time Series Forecasting.* Submitted to ICLR 2027 (under review).

Co-author papers

- Terrance G. Y. Corey, **Jingxin Zhang**, Christian Hill, Sergei N. Yurchenko, Jonathan Tennyson*. *A formal update of the ExoMol Database Collection*. RASTI, rzag081 (2026).
- Qing-He Ni, Rujia Wang, Tianyang Xie, **Jingxin Zhang**, Christian Hill, Sergei N. Yurchenko, Jonathan Tennyson*. *ExoAtom: a database of atomic spectra in ExoMol format*. RASTI, **5**, 1–23 (2026).
- Georgi B. Mitev, Marco Pezzella, Charles A. Bowesman, **Jingxin Zhang**, Sergei N. Yurchenko, Jonathan Tennyson*. *ExoMol photodissociation cross-sections – II. Continuum absorption and pre-dissociation spectra for the hydroxyl radical*. MNRAS, **539**(4):3732–3740 (2025).
- Georgi B. Mitev, Charles A. Bowesman, **Jingxin Zhang**, Sergei N. Yurchenko, Jonathan Tennyson*. *ExoMol line lists – LXI. A trihybrid line list for rovibronic transitions of the hydroxyl radical (OH)*. MNRAS, **536**(4):3401–3420 (2025).

- Sergei N. Yurchenko*, Charles A. Bowesman, Ryan P. Brady, Elizabeth R. Guest, Kyriaki Kefala, Georgi B. Mitev, Alec Owens, Armando N. Perri, Marco Pezzella, Oleksiy Smola, Andrei Sokolov, **Jingxin Zhang**, Jonathan Tennyson*. *ExoMol line lists – LX. Molecular line list for the ammonia isotopologue $^{15}NH_3$*. MNRAS, **533**(3):3442–3456 (2024).
- Jonathan Tennyson*, Sergei N. Yurchenko, **Jingxin Zhang**, et al. *The 2024 Release of the Exomol Database: Molecular Line Lists for Exoplanet and Other Hot Atmospheres*. JQSRT, **326**:109083 (2024).
- Jonathan Tennyson*, Marco Pezzella, **Jingxin Zhang**, Sergei N. Yurchenko. *Data structures for photoabsorption within the ExoMol project*. RASTI, **2**, 231–237 (2023).

GRANTS

Young Investigator Innovation Grant (Principal Investigator) — Dec 2025 – Dec 2027
Institute of Mechanics, Chinese Academy of Sciences — *Grant No. 2025QN22 | CNY 100 000*

- Deep learning based intelligent molecular and atomic identification from multi-band radiative spectra of high-temperature flow fields.

HONORS AND AWARDS

Jon Hougen Travel Award — Aug 2023
The 28th Colloquium on High-Resolution Molecular Spectroscopy (HRMS 2023) — *Dijon, France*

SCIENTIFIC CONSULTATION

International Atomic Energy Agency — May 2024
Invited Participant, Consultancy Meeting — *Atomic and Molecular Data Unit, Vienna, Austria*

- Invited to the Consultancy Meeting "Accurate Molecular Spectra for High-Temperature Applications" to contribute expertise in molecular spectroscopic databases, data interoperability, and standardisation, drawing on work on the ExoMolHR high-resolution spectroscopic platform.

SELECTED CONFERENCES

Oral presentations

ICAMDATA 2024 — Sep 2024
The ExoMol Database: developments and the 2024 data release — *Lanzhou, China*

HRMS 2024 — Sep 2024
ExoMolHR: a relational database of empirical high-resolution molecular spectra — *Bologna, Italy*

London Exoplanet Day — Dec 2023
The ExoMol database: developments and the 2024 data release — *London, UK*

Poster presentations

Exoplanets 5 — Jun 2024
The ExoMol Database: developments and the 2024 data release — *Leiden, Netherlands*

HRMS 2023 — Aug 2023
Development of the ExoMol database — *Dijon, France*

UKEXOM 2023 — Aug 2023
Development of the ExoMol database — *London, UK*

Spectroscopy of exoplanets at high resolution — Feb 2023
New features in the ExoMol database — *Northampton, UK*

TEACHING AND SUPERVISION EXPERIENCE

Co-supervisor, University College London — Oct 2022 – May 2025
Co-supervised four M.Sc. final research projects in Scientific and Data Intensive Computing, focusing on data infrastructure validation and automated computational workflows.

Postgraduate Teaching Assistant (PGTA), University College London — Jan 2024 – Jun 2024
Teaching demonstrator for PHAS0024 Statistical Physics of Matter.

TECHNICAL SKILLS

Programming & Tools	Python, Fortran, Bash/Shell scripting, Git, SQL, basic C++
Spectroscopy	Line lists, line intensities, line profiles, cross sections, LTE and non-LTE
Scientific Computing	NumPy, SciPy, Pandas, Dask, Numba, NumExpr, parallel and GPU workflows
Machine Learning	PyTorch, TensorFlow, Scikit-learn
Astronomical Applications	High-resolution spectroscopy, exoplanet spectra, molecular detectability
Research Methods	Spectral modelling, uncertainty filtering, data validation, reproducible workflows
Software	Adobe Creative Cloud (Premiere Pro, After Effects, Photoshop)
Interests	Piano, Traditional Ink Painting, Video Editing, Travelling